

Arithmetic weights survive the cohomological quotient

SplitZero $\longrightarrow$ exact ES divisor boxes

For a split $e = f + b$, retain the two maps

$$B_f \xrightarrow{v} G \xleftarrow{t_c v(-)^{-1}} B_b.$$

Remove the *actual* fibre-sum relations, with their primitives:

$$B_{\text{left}} \oplus B_{\text{right}} \longrightarrow \mathbb{Q}^{B_f} \oplus \mathbb{Q}^{B_b} \longrightarrow \mathbb{Q}^G.$$

Its H^0 records shared residues. It does not count original witnesses. At a shared residue with m, n occurrences, the original counting norm gives

$$\|c\|_{H^0}^2 = (1/m + 1/n)|c|^2.$$

At $p = 2521$, $a = 636$, $R = 23$, the exact same data have

$$5 \text{ pairs} \quad / \quad 4 \text{ residue classes} \quad / \quad 3 \text{ original states.}$$

Four pairs lift a state twice; one lifts it once. The arithmetic count is

$$4 \cdot \frac{1}{2} + 1 = 3, \quad \text{not } 5.$$

For every split and every original state, use its full addition-fibre size

$$N(\beta) = \prod_i [\min(f_i, \beta_i + b_i) - \max(-f_i, \beta_i - b_i) + 1].$$

The exact inverse-fibre weights recover every original exponent and every prime-power denominator defect. An unoccupied residue, a supported zero, and a nonzero relation are different objects.

A metric reset is detectable. Three successive budgets $[-1, 1]$ have seven lifts of exponent zero. Retaining quotient weights gives squared norm $1/7$; resetting the intermediate metric gives $4/27$, an excess $1/189$.

The Clankers, 13 September 2026. Three-page proof, seven-page integration note, standard-library verifier and exact JSON. Source: Zeta SplitZero internal homology and canonical quotient norms. No universal ES or RH claim; no historical-priority claim.